

# ISMAYIL YUSIFLI

Computer Science Student | AI/ML Engineering | Computer Vision

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## PROFILE

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Computer Science student at UFAZ focused on AI/ML engineering and applied computer vision. Co-Founder of Guardian-X, contributing to the computer-vision pipeline behind AZira using PyTorch, FastViT, ONNX, and TensorFlow Lite. Experience also includes classical machine learning, model evaluation, neural networks from scratch, custom computer-vision datasets, and deployment-oriented AI engineering.

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## EDUCATION

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**French-Azerbaijani University (UFAZ) — BSc in Computer Science**

**2023 — 2027**

Cumulative average: **14.485/20** | Official equivalent academic standing: **98/100**

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## EXPERIENCE

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**Co-Founder — Guardian-X | AI/ML Engineering**

**Aug 2025 — Present** Baku, Azerbaijan | Hybrid

- Contribute to the AI/ML engineering behind **AZira**, an AI-powered pest detection application publicly available on Google Play.
- Developed and evaluated a **FastViT-T12 image-classification pipeline** for pest recognition across **15 classes** using approximately **18,000 images**.
- Worked on dataset preparation, corrupted-image filtering, class mapping, transfer learning, validation, testing, and generalization checks across train, validation, and test splits.
- Converted the trained PyTorch model through **ONNX and TensorFlow Lite** for mobile integration and investigated model-conversion issues during deployment preparation.
- Tested mobile conversion variants, identified unreliable float16 behaviour, and retained the working float32 model for more reliable integration.
- Contributed to AI integration planning and coordination between the model pipeline and the broader mobile product.

**Machine Learning Intern — FlyRank AI**

**Jul 2026 — Present** Remote | Self-paced, project-based

- Built and executed an end-to-end machine-learning pipeline on approximately **30,000 records**, comparing a rule-based ranking approach with a Random Forest model.
- The Random Forest achieved **Precision@50 of 0.74 versus 0.24 for the rule-based baseline**.
- Investigated feature selection, decision-tree depth, feature importance, target leakage, generalization, and validation methodology.
- Maintained executable notebooks and reproducible experiments, validating changes by rerunning pipelines and checking outputs.

**C/C++ Developer Intern — Intern Intelligence**

**Feb 2025 — Mar 2025** Remote

- Built an advanced mathematical-functions library using C and C++.
  - Developed a multithreaded task scheduler for managing and executing concurrent tasks.
  - Worked with data structures, algorithms, debugging, multithreading, CMake, and testing.
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## TECHNICAL SKILLS

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**Machine Learning & AI:** PyTorch, scikit-learn, NumPy, Pandas, OpenCV, CNNs, FastViT, image classification, transfer learning, model evaluation, experiment design

**Programming:** Python, C++, C, Java, SQL

**Model Deployment & Engineering:** ONNX, TensorFlow Lite, Docker, Git, Flask, REST APIs, debugging, testing

**AI Engineering Tools:** Codex, Claude Code, Gemini, DeepSeek; AI-assisted debugging, testing, and code review

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## SELECTED PROJECTS

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### From-Scratch Softmax Regression & Neural Network Classifier

Python, NumPy, Matplotlib

- Built softmax regression and a one-hidden-layer neural network from scratch with NumPy, implementing forward propagation, loss computation, and training logic without relying on high-level deep-learning frameworks.
- Implemented mini-batch training, validation checkpointing, reproducible experiment runners, and optimization using **SGD, Momentum, and Adam**.
- Ran repeated-seed and hidden-layer-capacity experiments across Gaussian, moons, and handwritten-digit datasets to compare linear and nonlinear models using accuracy and cross-entropy loss.

### Azerbaijani Handwritten Character Recognition

Python, CNN, OpenCV, Flask

- Built an end-to-end computer-vision system for recognizing handwritten Azerbaijani characters using a custom collected dataset, covering dataset preparation, preprocessing, model training, evaluation, and inference.
- Developed an OpenCV-based preprocessing workflow and trained a CNN classifier, evaluating predictions using classification metrics.
- Integrated the trained model into a Flask-based interactive prediction demo, turning the model into a usable web application.

### CourseFlow — AI Coursework Execution Planner

Gemini, Google GenAI SDK, TypeScript, Node.js, Express

- Built an AI-powered coursework execution planner that converts uploaded coursework into evidence-backed requirements, decomposed tasks, workload estimates, and an execution schedule.
  - Combined structured LLM outputs with deterministic dependency scheduling, workload-feasibility checks, and constraint-aware replanning.
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## AI & ENGINEERING TRAINING

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- **AI Engineering Program** — AI Academy, Ongoing
  - **Math4AI Track** — AI Academy, Completed
  - **Claude with the Anthropic API** — Anthropic Academy
  - **AI Fluency: Framework & Foundations** — Anthropic Academy
  - **Certificate of Proficiency — Java & Back-End Web Development** — Code Star
  - **Mendix Rapid Developer Certification** — Mendix
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## ACHIEVEMENTS

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- **3rd Place** — Siemens Sustainable Design Hackathon, Guardian-X Team
  - **Honorable Mention** — 2025 ICPC Azerbaijan Qualification
  - **PASHA Hackathon 6.0** — Built a retail waste-management solution with a team
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## LANGUAGES

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**Azerbaijani:** Native • **English:** Advanced / C1 • **Turkish:** Intermediate • **French:** Elementary / A2